

# SIDDHARTH KUMAR TIRKEY

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Cisco Certified DevNet Specialist | Cisco Certified Network Professional

## PROFESSIONAL SUMMARY

Network Automation Engineer with 4.5 years of experience at Cisco Systems, specialising in large-scale network fabric automation, operational tooling, and distributed system deployments. Strong hands-on expertise with Layer 2-3 protocols (BGP, EVPN-VXLAN, SDA) at an operational and validation level, and with Layer 7 API integrations (Cisco Catalyst Center REST APIs). Experienced in Python-based framework development, CI/CD pipeline engineering, and infrastructure-as-code. Industry presenter at Cisco Live (2x) and COMSNETS.

## WORK EXPERIENCE

### Consulting Engineer | Cisco Systems India Pvt. Ltd.

Aug 2021 – Present

- Architected and led development of a configuration automation platform for BGP EVPN-VXLAN fabrics using Python, Ansible, Jinja2, and NetBox, reducing deployment effort by 80% across large-scale enterprise environments.
- Developed an in-house Python-based vendor-agnostic network state analysis framework for gathering, parsing, and validating device state across heterogeneous environments, accelerating network design and redesign cycles by 80%.
- Designed and deployed GitLab CI/CD pipelines with Terraform for automated, change-driven network provisioning, enabling repeatable infrastructure-as-code workflows aligned with DevOps and Agile practices.
- Engineered integrations with Cisco Catalyst Center REST APIs and SDA assurance APIs (Layer 4-7), improving operational visibility, monitoring, and automated troubleshooting across distributed network fabrics.
- Delivered on-site deployments for large-scale distributed environments including cruise ship networks, applying automated validation, state checks, and real-time troubleshooting under mission-critical conditions.
- Built RAG and LLM-powered internal chatbots (LangChain, LlamaIndex, pydantic-ai) to accelerate knowledge retrieval and internal engineering workflows.
- Speaker and hands-on-lab host at Cisco Live (2x) and COMSNETS, presenting on SDA automation, Catalyst Center APIs, and zero-touch provisioning to senior technical audiences.

### Technical Undergraduate Intern | Cisco Systems India Pvt. Ltd.

Jan 2021 – Jun 2021

- Designed a Python-based tool to detect and audit feature configurations across switch inventories, enabling automated compliance checks at scale.

## TECHNICAL SKILLS

|               |                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|
| Languages     | Python (advanced), Bash, SQL, C (intermediate)                                                               |
| Networking    | BGP, EVPN-VXLAN, Cisco SDA, Layer 2-3 routing/switching, REST APIs, TCP/IP, HTTP, UDP, Cisco Catalyst Center |
| Automation    | Ansible, Jinja2, NetBox, Robot Framework, FastAPI                                                            |
| DevOps / IaC  | Terraform, GitLab CI/CD, Git, Agile, Test-Driven Development                                                 |
| Cloud & Infra | Containers (Docker), Kubernetes (fundamentals)                                                               |
| AI / LLM      | RAG, LangChain, LlamaIndex, pydantic-ai agents                                                               |

## EDUCATION

B.Tech in Computer Science and Engineering

2017 – 2021

## ACHIEVEMENTS

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- Winner, Accenture Hack the Future Gen AI Hackathon 2025, built a multi-agentic system correlating data across sources to deliver actionable, data-driven recommendations to farmers.